

AMAZON FRESH

Analytics

SQL Data Analysis Project

Turning e-commerce data into structured,
decision-ready business insights.

PROJECT PRESENTATION

Presented by

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Customers • Orders • Products • Reviews • Suppliers

AMAZON

RETAIL ANALYTICS

SQL

Data Layer

BI

Dashboard

360°

Insights

INTRODUCTION

PROJECT PURPOSE

Amazon Fresh Analytics is a SQL-based data analysis project designed to analyze customer behavior and product sales.

ANALYTICAL FOCUS

The analysis focuses on customer spending, product sales, ratings, and reviews.

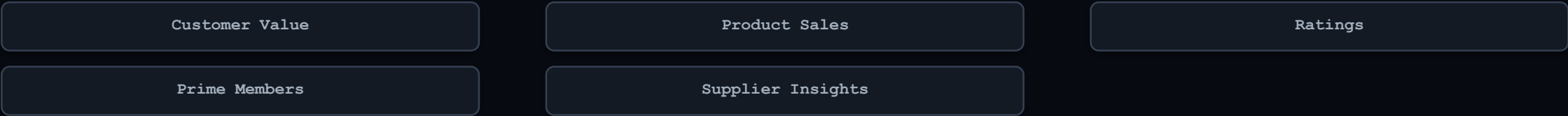
SQL-DRIVEN SOLUTION

The project uses SQL queries to retrieve, clean, transform and analyze data from various sources.

BUSINESS OUTCOME

The goal is to generate useful insights that can support better business decisions.

Core business themes



PROBLEM STATEMENT

Why structured analytics is needed

01

DATA COMPLEXITY

Large e-commerce datasets contain customer, order, product and supplier information that can be difficult to analyze manually.

02

BUSINESS QUESTIONS

Businesses need to identify high-value customers, top-selling products and frequently ordered categories.

03

DATA QUALITY

Duplicate names, invalid ages and invalid ratings can affect analysis.

04

PROJECT GOAL

Use SQL to organize data, maintain data integrity and extract actionable business insights.

APPROACH

01

DATABASE

Create and retrieve data from the Amazon Fresh da

02

QUERY

Use WHERE, GROUP BY, HAVING and ORDER BY to filte

03

CALCULATE

Apply SUM, COUNT and AVG for business calculations.

04

COMBINE

Use JOINS and subqueries to answer complex busine

05

RANK

Use RANK() to rank customers by total spending.

06

NORMALIZE

Normalize product category information toward 3NF using

SQL → Clean Structure → Analysis → Ranking → Business Insight

RESULTS

CUSTOMER VALUE

High-value customers spending more than ₹5,000 were identified

PRODUCT PERFORMANCE

The top 3 products by sales revenue were identified using Order Details

CATEGORY DEMAND

The top 3 most frequently ordered product categories were identified

PRIME ANALYSIS

Cities with the highest concentration of Prime members were analyzed

ORDER ANALYSIS

Orders and order details were joined to analyze order-level product

DATA INTEGRITY

Primary key, unique, CHECK, DEFAULT and foreign-key constraints implemented

PROJECT EVALUATION METRICS

DATA INTEGRITY

Primary key, unique, CHECK, DEFAULT and foreign

QUERY ACCURACY

Correctness of filtering, aggregation, joins, su

BUSINESS INSIGHT

Identify high-value customers, top products and popul

DATA ORGANIZATION

Normalization of product category information to

QUERY EFFICIENCY

Appropriate GROUP BY, ORDER BY, LIMIT and JOIN c

COMPLETENESS

Coverage of customer, order, product, review and supp

TECHNICAL STACK

Core SQL capabilities demonstrated



MySQL / SQL



DDL and DML



Primary Key and Foreign Key



CHECK / UNIQUE / DEFAULT



JOINS



Subqueries and Nested Queries



Aggregate Functions



GROUP BY / HAVING / ORDER BY



Window Function - RANK()



Database Normalization - 3NF

Focus: reliable queries + structured data + actionable analysis

DATASET

CUSTOMERS

Customer details, city, age and Prime membership.

ORDERS

Order ID, customer ID, order date, order amount, delivery fee and discount.

ORDER DETAILS

Product ID, quantity, unit price and discount for each order.

PRODUCTS

Product name, category, subcategory, price, stock quantity and supplier.

REVIEWS

Product ratings used for rating analysis.

SUPPLIERS

Supplier information used for supplier-related analysis.

Database: Amazon

6 core entities support customer, transaction, product and review analysis.

Professional Power BI Dashboard

Presented by **Kavinprasath.R**

Executive overview • KPI monitoring • interactive business analysis



CONCLUSION

What the project demonstrates

01

SQL can transform raw e-commerce data into useful business insights.

02

SQL constraints and normalization improve data quality and database structure.

03

Joins, aggregations, subqueries and window functions support detailed business analysis.

04

The project identifies valuable customers, top products, popular categories and Prime-member concentrations.

05

Overall, the analysis provides a strong foundation for data-driven decision-making in an e-commerce environment.

THANK YOU • Questions & Discussion

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